

## **Project 1 - Credibility Scoring for Sources**

### **Introduction**

This project focuses on improving a system that gives a credibility score to a web source. The score goes from 0.0 to 1.0, where a higher value represents a source that is considered more credible, and the program also explains which signals affected the result.

One challenge with this type of system is deciding how to evaluate a source that is not already known. Esteves et al. (2018) discuss a similar problem with manually created lists of online sources, since these lists do not scale well when new or unknown websites appear. Their work uses sources reputation signals to calculate a credibility factor, which is related to the idea used in this project of combining different signals instead of depending only on a fixed list of domains.

Preprints create another challenge because they are academic sources, but they are different from peer-reviewed journal articles. Soderberg et al. (2020) study how researchers evaluate the credibility of preprints and found that different cues can affect these judgments. Their work also explains that preprints do not have the same journal reputation, selection, and peer-review signals normally associated with published journal articles.

There is also an important difference between evaluating a source and deciding whether the information itself is true. Torabi Asr and Taboada (2018) discuss the difference between source reputation and content veracity, which is important for this project because the system estimates the credibility of a source, but it does not verify every claim found on the page.

The starting version of the project already used more than one signal, with a rule-based layer that looked at the URL and an optional Claude layer that provided a second credibility score. The rules considered information such as the domain, top-level domain, HTTPS, and DOI patterns, while the original final score combined 60% rules with 40% Claude.

After running the baseline evaluation, some problems became clear. The rules worked well for many domains already recognized by the program, but some credible sources that were not in the domain table received low scores. Preprints had the opposite problem, since academic signals could make them score too high even though the papers had not necessarily completed peer review.

Instead of adding more websites to the domain table, the project focused on the problems found during the evaluation. The main changes were preprint detection, additional signals from the URL path, and different weights for combining the rule-based score with Claude.

## Baseline

The rule-based system starts by looking at the domain, where some known sources already have predefined credibility scores, including Nature, Science, NEJM, PubMed, and Reuters. The program then looks for other signals in the URL that can increase or decrease the score.

For example, `.gov` and `.edu` receive more positive values than extensions such as `.info` and `.xyz` while HTTPS and DOI patterns can also provide positive signals. This approach is simple and fast, and it can still produce a credibility score without using the Claude API.

The first run of `evaluate.py` showed where the rules were having trouble. JAMA had an expected score of 0.93 but received only 0.52, while WHO received 0.52 instead of 0.88. bioRxiv showed a different type of error, with an expected score of 0.50 but a rule-based score of 0.82.

The bioRxiv result was useful because it showed that the problem was not only credible sources receiving scores that were too low. Academic signals in the URL could also push some scores too high, especially when the system did not distinguish between a preprint and a peer-reviewed publication,

The original rules had an MAE of 0.142, band accuracy of 66.7%, and a worst error of 0.410. Adding Claude with the original 60/40 weights improved the results, bringing MAE to 0.086, band accuracy to 83.3%, and worst error to 0.230.

## Improvements

To help compare different options, a separate Python script called `tune_weights.py` was created. The script was used for the three main experiments in the project: testing the

preprint penalty, testing the URL path bonus, and comparing different weights for the rule-based and Claude scores. It calculated the MAE for different values, which made it easier to compare how each option affected the results. These results were used as a guide when deciding which values to use in `credibility.py`.

## Preprints

Preprints were one of the first problems found in the baseline, especially because bioRxiv received 0.82 when its expected score was 0.50. bioRxiv is still an academic source, but papers posted there may not have completed peer review, and the original rules were giving too much credit to the academic signals in the URL.

Using `tune_weights.py`, the preprint penalty was tested from 0.00 to 0.50. Values from 0.12 through 0.22 produced the same lowest MAE of 0.050 for the preprint examples. Since several values gave the same result, 0.12 was selected because it was the smallest penalty in that range.

Based on this result, a preprint signal was added for arXiv and bioRxiv, subtracting 0.12 from the rule-based score when one of these platforms is detected. The DOI bonus is also disabled for these preprints, since a DOI can be a positive academic signal but does not mean that a paper has completed peer review.

## URL Path

The next problem appeared in sources such as JAMA, WHO, and IMF. The original rules did not recognize enough positive information from these URLs, but adding the three domains directly to the known-domain table seemed too specific and would not help with other unknown sources.

Their URL paths contained terms such as `journals`, `fact-sheets`, and `publications`, so these words were tested as positive signals. They do not prove that a page is credible, but they can provide additional information about the type of page when the domain itself is not already recognized.

The URL path bonus was also tested with `tune_weights.py`. Testing the three examples suggested a bonus of 0.36, but the experiment used only three URLs, so using the best value from such a small sample could make the rule too specific to those examples. A more conservative bonus of 0.10 was selected to give the URL path some influence without

allowing a single word in the path to change the score too much. The 0.10 bonus was then evaluated as part of the rule-based system across all 24 URLs.

After the preprint and URL path changes, the rules-only MAE decreased from 0.142 to 0.115, band accuracy increased from 66.7% to 79.2%, and worst error decreased from 0.410 to 0.310.

## **Claude Weight**

The last experiment tested the balance between the rule-based score and Claude. The original program used 60% rules and 40% Claude, but these weights were not selected from an evaluation.

Using the same `tune_weights.py` script, different combinations were tested in steps of 10%. The first tuning run showed that 20% rules and 80% Claude produced the lowest MAE at 0.054. Since Claude return slightly different scores between API calls, the experiment was run again, and the 20/80 combination produced the lowest MAE again, this time at 0.052.

A 30/70 combination was also checked during the final evaluation, but 20/80 still produced a lower MAE and lower worst error. Based on these results 20% rules and 80% Claude were used as the final weights.

## **Results**

The same 24 labeled URLs in `evaluate.py` were used to compare the different versions of the system. The examples include scientific journals, government sources, news organizations, documentation, preprints, personal blogs, social media, satire, and suspicious websites.

Three metrics were used for the comparison. MAE measures the average distance between the predicted score and the expected score, so a lower value means that the predictions are closer to the expected values. Band accuracy measures how often a source is placed in the expected HIGH, MEDIUM, or LOW category, while worst error shows the largest individual difference found in the evaluation.

| Version                       | MAE   | Band Accuracy | Worst Error |
|-------------------------------|-------|---------------|-------------|
| Original rules                | 0.142 | 66.7 %        | 0.410       |
| Original rules + Claude       | 0.086 | 83.3 %        | 0.230       |
| Improved rules                | 0.115 | 79.2 %        | 0.310       |
| Improved rules + Claude 60/40 | 0.072 | 95.8 %        | 0.170       |
| Final rules + Claude 20/80    | 0.049 | 91.7 %        | 0.130       |

The improvements to the rules reduced the MAE and worst error while also increasing band accuracy. The biggest change appeared in the final hybrid version, where MAE dropped from the original 0.086 to 0.049 and the worst error dropped from 0.230 to 0.130.

One result was a little unexpected. The improved 60/40 version had better band accuracy, 95.8% compared with 91.7% for 20/80. On the other hand, the 20/80 version had the better MAE and the lower worst error, meaning that its numerical scores were closer to the expected scores overall. Since MAE was the main metric used during weight tuning, 20/80 was kept as the final combination.

Three additional tests were added to `test_credibility.py` to validate the main improvements: preprint detection, positive URL path detection, and DOI handling for preprints. After adding these tests, the complete test suite was run again, and all 24 tests passed.

## Limitations

The evaluation dataset contains only 24 labeled URLs, so the results show how the different versions compare in this project but do not guarantee the same performance on a much larger group of websites. The URL path experiment made this limitation clear, since a bonus of 0.36 looked best on three examples but there was not enough evidence to give one path word that much influence on the score.

Another limitation comes from what the rule-based layer analyzes. The system looks at the URL rather than the full content of the page, so it does not check the author, references, publication date, corrections, or evidence behind a claim. A credible website can still contain weak information, just as an unknown website can contain useful information.

Claude adds a different type of uncertainty because its score can change slightly between calls. This appeared during the weight tuning, where the MAE changed slightly between the first and second runs even though the same URLs were used.

The arXiv test showed an interesting limitation of the hybrid approach. The rule layer correctly detected the URL as a preprint and applied the penalty, but Claude still gave it a score of 0.80. Claude recognized that arXiv is a preprint platform, but it also recognized the specific paper, "Attention Is All You Need", and considered that it was later peer-reviewed. In some ways, this is what Claude is expected to do, since it can consider more context than the rules alone. At the same time, it raises a question about the score. Should Claude evaluate the paper using what is known about it today, including its later peer review, or should the score reflect only the status of the paper when it was originally posted as a preprint? This is something that could be explored in future improvements to the system.

Finally, the hybrid version depends on the Anthropic API. If the API key is unavailable or the call fails, the program can still use the rule-based layer, but the current evaluation shows that the rules-only version has lower performance.

## Conclusion

The project started with a rule-based credibility scorer and an optional Claude layer, then focused on problems found during the baseline evaluation instead of adding more domains to the known-domain table.

Preprint detection helped with sources that were receiving too much credit from academic signals, while URL path signals provided more information for credible sources that were not already recognized. Testing the rule and Claude weight also showed that the original 60/40 split was not the best combination when MAE was used as the main metric.

The final hybrid version reached an MAE of 0.049 compared with 0.086 for the original hybrid system, while the worst error decreased from 0.230 to 0.130. The results improved overall, but the higher band accuracy of the 60/40 version showed that the lowest MAE does not always produce the highest band accuracy.

The main limitation is still the information available to the system. URL signals can help estimate the credibility of a source, but they cannot prove that the content on a page is correct, while Claude can provide useful context but also introduces some variation between calls. The final system improved the measured results of this project but there is still room to make the credibility assessment more complete.

## Future Work

Future work could evaluate the system with a larger and more diverse dataset. With more labeled URLs, the rule values and the rule/Claude weights could be tested with separate training and evaluation data, reducing the risk of overfitting to the current 24 examples.

Another direction would be to explore the question found during the arXiv test. Future experiments could compare whether Claude should evaluate a source using information known today or only information that was available when the content was originally published. This could help define more clearly what the credibility score is intended to represent.

## References

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